Chapter 6

Special Features of the EPIC and APEX Modeling Package and Procedures for Parameterization, Calibration, Validation, and Applications

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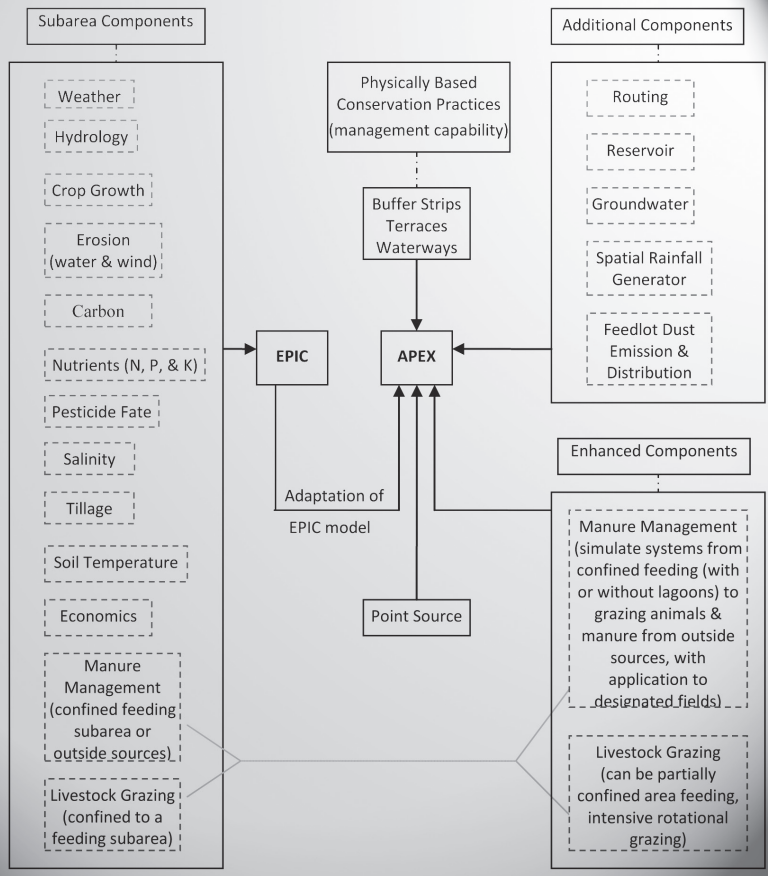
Abstract

Numerous hydrologic and environmental models, with a range of different levels of approach and complexity, have been developed to strengthen and complement field studies. The Environmental Policy Impact Climate (EPIC) and the Agricultural Policy Environmental eXtender (APEX) models are flexible and dynamic tools that were developed to evaluate a wide array of management strategies applied to crop, pasture, and grazing lands. They are capable of estimating long-term sustainability of land management in respect to erosion (wind, sheet, and channel), economics, water supply, water quality, soil quality, plant competition, weather, and pests for cropland, as well as grazing and pasture land. This chapter describes the main features of the EPIC and APEX models and reports on the parameterization, calibration, and validation of the models. As a case study, the APEX model was applied to the North Bosque watershed in central Texas, where the major land use/land cover is native pastures/range and improved pasture, with dairies and milk sheds. Influential model parameters were adjusted during the model calibration period on the basis of the monthly stream flow, sediment yield, and nutrient losses at the Hico monitoring station. The model performance was reasonable, and this application shows that the APEX model is a powerful tool for simulating feedlot water quality.

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Subarea Components
Weather
Hydrology
Crop Growth
Erosion (water & wind)
Carbon
Nutrients (N, P, & K)
Pesticide Fate
Salinity
Tillage
Soil Temperature
Economics
Manure Management (confined feeding subarea or outside sources)
Livestock Grazing (confined to a feeding subarea)
EPIC
Physically Based Conservation Practices (management capability)
Buffer Strips
Terraces
Waterways
APEX
Adaptation of EPIC model
Point Source
Additional Components
Routing
Reservoir
Groundwater
Spatial Rainfall Generator
Feedlot Dust Emission & Distribution
Enhanced Components
Manure Management (simulate systems from confined feeding (with or without lagoons) to grazing animals & manure from outside sources, with application to designated fields)
Livestock Grazing (can be partially confined area feeding, intensive rotational grazing)

The Environmental Policy Impact Climate (EPIC) model (Williams et al., 1984; Williams, 1990, 1995) was developed in the early 1980s to assess the effect of soil erosion on soil productivity. It can be used to determine the effect of management strategies on agricultural production and soil and water resources. The drainage area considered by EPIC is generally a field-sized area of up to 100 ha, where weather, soils, and management systems are assumed to be homogeneous. The major components in EPIC are hydrology, erosion–sedimentation, nutrient cycling, pesticide fate, plant growth, soil temperature, tillage, economics, and plant environment control. It also includes a weather generator (Fig. 6–1). The Agricultural Policy Environmental eXtender (APEX) model (Williams et al., 1995; Williams, 2002; Williams and Izaurralde, 2006) is an extended and expanded version of the EPIC model. It was developed to facilitate multiple fields' scenarios in whole-farm or watershed management. The farm or watershed may be subdivided based on fields, soil types, landscape positions, or subwatersheds. Therefore, APEX can provide a consistent approach for evaluating various land management strategies at scales ranging from field to farm to watershed. Both EPIC and APEX run on a daily time-step. In APEX modeling, the individual field simulation uses the functions originally developed in EPIC (Fig. 6–1). In addition to EPIC functions, APEX has components for routing water, sediment, nutrients, and pesticides across landscapes through channels, floodplains, and reservoirs to the watershed outlet. The APEX groundwater component partitions flow between deep percolation and return flow. APEX also has a grazing component that provides flexibility to simulate confined or partially confined feeding area, intensive rotational grazing, cropland grazing after harvest, and other options (Fig. 6–1). Both EPIC and APEX models have been widely applied (Gassman et al., 2005, 2010).

The EPIC and APEX models require a relatively large number of input parameters. Like any other model, proper parameterization of EPIC and APEX is essential for making use of their modeling capabilities. For example, while the EPIC model was used for crop rotations in Southern France (Cabelguenne et al., 1990), the crop parameters including biomass energy ratio (WA), harvest index (HI), optimum air temperature (T_0), base temperature (T_b), and maximum leaf area index (LAI_{mx}) were adjusted to obtain a more satisfactory simulation of the seasonal development of leaf area index and grain yield. Gaiser et al. (2010)

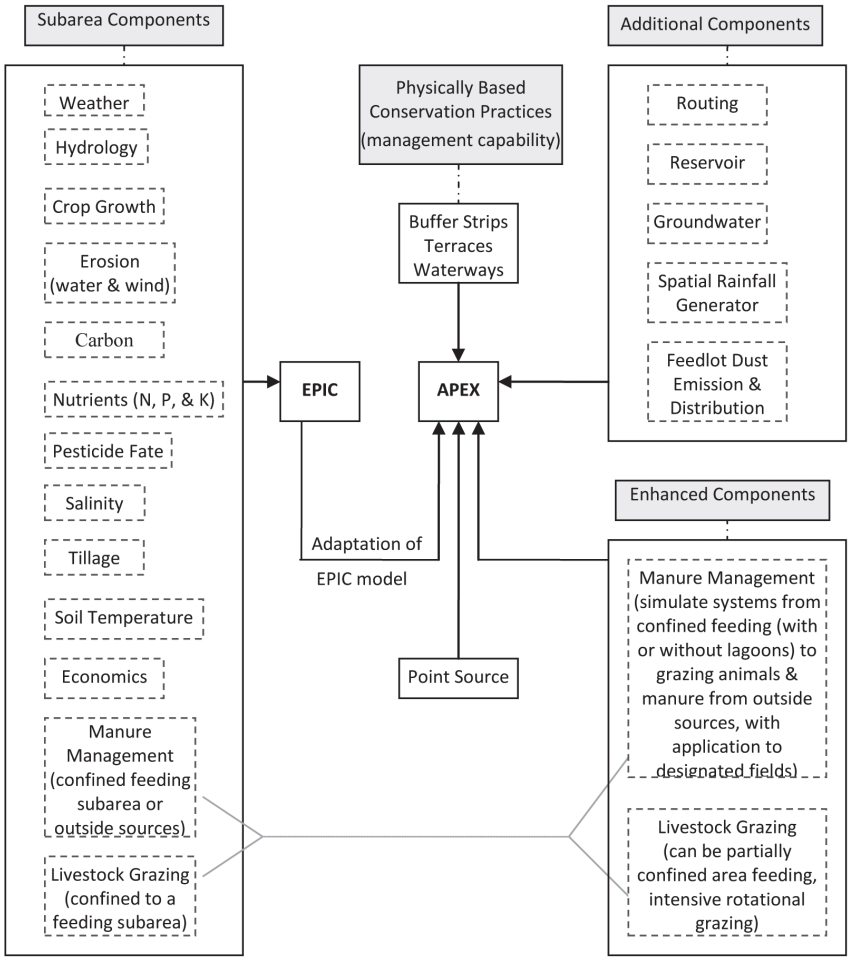


Fig. 6–1. The EPIC and APEX models.

indicated that crop growth models in tropical regions require extended testing under a range of climatic, soil, and management conditions and they used multi-location data to calibrate and validate the EPIC model for maize production in western Africa and northeastern Brazil. Wang et al. (2005) tested the EPIC crop growth and soil organic C components and conducted a sensitivity analysis for nine EPIC parameters using a variance-based sensitivity analysis method—the extended Fourier amplitude sensitivity test (FAST) (Saltelli et al., 1999, 2000). In Wang et al. (2006c), the sensitivities of 15 key APEX parameters involving hydrology, water and wind erosion, nutrient losses, soil organic C, and crop growth were analyzed for 159 locations across the conterminous United States. The anal-

ysis covers a wide range of soil types and climatic conditions, includes cropping systems of corn (*Zea mays* L.), soybean [*Glycine max* (L.) Merr.], and wheat (*Triticum aestivum* L.), and tillage systems of no-till, mulch, and conventional till. Yin et al. (2009) performed a sensitivity analysis for 13 key parameters affecting surface runoff and sediment loss using the FAST method before the model testing phase. Collectively, the USDA-NRCS curve number (CN), curve number index coefficient (PARM42), and the Hargreaves potential evapotranspiration (PET) equation (Hargreaves and Samani, 1985) exponent (PARM34) (if the Hargreaves method is used) are influential for runoff- and water-related output variables, such as soil loss by water and N and P losses in runoff. The revised universal soil loss equation (RUSLE) C factor coefficients (PARM46 and PARM47) are influential for erosion, sediment yield, and N and P losses in sediment. The available soil water capacity (the difference of soil water contents at field capacity and wilting point), potential heat units (PHU), biomass/energy ratio (WA), and harvest index are influential for the crop growth component (Table 6–1). The microbial decay rate coefficient and fraction of humus in passive pool are influential for the soil organic C component. The nitrate leaching ratio (nitrate concentration in surface runoff/nitrate concentration in percolate) and soluble P sorption coefficient (P concentration in sediment divided by that of the water) are influential to P loss (Table 6–1). Although numerous studies have been conducted using the EPIC and APEX models, as reviewed by Gassman et al. (2005; 2010), there is a need to summarize the step by step procedures for the models' parameter estimation and calibration. The objectives of this chapter are to describe the special features of the EPIC and APEX modeling package, to discuss parameterization, calibration and validation issues, and to demonstrate an APEX model's application.

Special Features of the EPIC and APEX Modeling Package

The EPIC and APEX models have been actively maintained since their inception. The models are refined, expanded, and improved constantly (Williams et al., 2008), a process mainly driven by various users' interaction, feedback, and interests for specific applications. Consequently, the models have a wide range of applications (Gassman et al., 2005, 2010) and also contain multiple options for simulating processes, such as hydrology and soil erosion–sedimentation (Table 6–2). Therefore, users can prefer one over the other depending on data availability and geographic or agricultural settings.

The models dynamically represent multiple soil properties, including depth and bulk density for each soil layer. The models remove eroded soil, attached organic nutrients, pesticides, and C from the soil profile as part of the emphasis

Table 6–1. Review of influential EPIC and APEX parameters.

Reference	Study location	Model component	Influential parameter†
Cabelguenne et al. (1990)	Southern France	crop grow	WA; HI; T_0; T_b; LAI_{mx}; RDMX; DLAI; DLAP1; DLAP2; RLAD; CNY; WSYF; BN1; BN2; BN3
Chung et al. (2001, 2002)	Lamberton, MN Nashua, IA	hydrology; N cycling	CN2; N fixation coefficient (PARM7)
Wang et al. (2005)	Plots in WI Arlington, Agricultural Research Station in south-central Wisconsin	crop grow; carbon	Soil water capacity (FC-FW); potential heat units (PHU); WA; HI; PARM70 (used to be PARM20 in old EPIC version); FHP
Wang et al. (2006a)	Zi-Fang-Gully watershed in northwestern China	hydrology; erosion/ sedimentation	soil saturated conductivity (SATC); floodplain saturated conductivity (FPSC); PARM42; PARM46; PARM47
Wang et al. (2006b)	Riesel, TX	hydrology; erosion–sedimentation; nutrients	CN2; PARM42; PARM46 (used to be PARM23 in old version); PEC; phosphorus sorption ratio (PSP)
Wang et al. (2006c)	159 locations in the United States	hydrology; crop grow; erosion–sedimentation; nutrients; carbon	PARM2; PARM5; PARM8; PARM11; PARM34; PARM42; PARM46; PARM52; PHU; UXP; FHP; return flow ratio (RFP)
Wang et al. (2008)	Treynor watersheds in Iowa	hydrology; carbon; erosion–sedimentation	CN2; PARM42; PARM46; PARM47; PARM70
Yin et al. (2009)	Middle Huaihe River watershed, China	hydrology; erosion–sedimentation	Peak runoff rate-rainfall energy adjustment factor (APM); CN2; PARM42; PEC
Wang et al. (2009)	Shoal Creek watershed, central Texas	hydrology; erosion–sedimentation	CN2; PARM18; PARM42; PARM46; RSRR
Gaiser et al. (2010)	Tropical subhumid West Africa and semiarid Brazil	crop grow	WA; HI, WSYF; LAI_{mx}; PPLP1; PPLP2; Aluminum tolerance index (ALT); Critical aeration factor (CAF)
Mudgal et al. (2010)	Goodwater Creek watershed in north central Missouri	hydrology; pesticide fate	PARM3; PARM5; PARM16; PARM17; PARM24; PARM34; PARM38; PARM42; PARM44; PARM63; PHLS

Continued.

Table 6–1. Continued.

Reference	Study location	Model component	Influential parameter†
Wang et al. (2010)	subwatershed within the Bosque River watershed in central Texas	hydrology; nutrients; erosion–sedimentation	CN2; PARM8; PARM14; PARM18; PARM19; PARM29; PARM31; PARM35; PARM42; PARM46; PARM59; PARM72

† BN1–3: normal fraction of N in crop biomass at emergence, mid-season, and maturity, respectively; CN2: Initial input of condition 2 curve number; CNY: normal fraction N in yield; DLAP1 & DLAP2: first and second point on leaf development curve, respectively; DLAI: fraction of growing season when LAI declines; HI: harvest index; LAI_{mx}: maximum leaf area index; PARM2: root growth soil strength; PARM3: water stress harvest index; PARM5: soil water lower limit in the top 0.5 m soil depth; PARM8: soluble P runoff coefficient; PARM11: moisture fraction required for seed germination; PARM14: nitrate leaching ratio, which is the ratio of nitrate concentration in surface runoff to that in the percolate; PARM16: expands CN retention parameter; PARM17: soil evaporation plant cover factor; PARM18: sediment routing exponent; PARM24: pesticide leaching ratio; PARM29: biological mixing efficiency simulates mixing in top soil by, e.g., earthworms; PARM31: maximum depth for biological mixing; PARM34: Hargreaves PET equation exponent; PARM35: denitrification soil-water threshold, which is the fraction of field capacity soil water storage to trigger denitrification; PARM38: water stress weighting coefficient; PARM42: NRCS curve number index coefficient; PARM44: upper limit of CN retention parameter; PARM46 & PARM47: RUSLE C factor coefficients; PARM52: exponential coefficient of tillage effect on residue decay rate; PARM59: P upward movement (by evaporation) coefficient; PARM63: pesticide loss coefficient; PARM70: microbial decay rate coefficient; PARM72: volatilization/nitrification partitioning coefficient; PEC: erosion control practice factor; PHLS: pesticide half-life in soils; PPLP1 & PPLP2: plant density_LAI parameters; RDMX: Maximum root depth; RLAD: leaf area index decline rate parameter; RSRR: average time to release flood storage through the rock dams in day; T_0: optimum air temperature; T_b: base temperature; WA: energy conversion ratio; WSYF: lower limit of harvest index; FHP: fraction of humus in passive pool; UXP: power parameter of modified exponential distribution of wind speed.

on erosion productivity. Soil hydraulic properties and bulk density are estimated from soil texture and soil organic C using the equation described in Rawls and Brakensiek (1985). The models also consider the effect of changes in atmospheric CO_2 on growth and transpiration and can incorporate trends in technology change through changes in the harvest index.

The models enable simulation of mixed stands with plant competition for light, water, and nutrients. An extensive array of tillage systems can be simulated. Tillage mixes soil layers, changes standing residue to flat residues, accelerates mineralization, and changes bulk density. The soil bulk density gradually increases after tillage based on the amount of water infiltrating through the corresponding layer. The difference in bulk density between a settled and tilled condition controls the effect of tillage in the mineralization rate. Both the soil bulk density settling and its effect on the mineralization rate are represented by empirically derived equations. The models enable wind erosion. Manure can also be eroded from feeding areas and manure application fields. Depending on the amount of manure cover of the soil the erosion varies from essentially all manure to a combination of manure and soil. Since manure is considered residue, a heavy cover in a feedlot may completely eliminate soil erosion but may create the potential for severe manure erosion. Soil erosion potential is also very low in manure applica-

Table 6–2. Major components and options available in the EPIC and APEX models.

Component	Method or principle
Weather	Weather generator options, if not input: First-order Markov chain model for precipitation (Nicks and Lane, 1989) Multivariate generation model for air temperature/solar radiation (Richardson, 1981) Daily wind speed distribution and dominant direction model (Richardson and Wright, 1984) Relative humidity model for daily average relative humidity
Hydrology	
Potential evapotranspiration	Penman–Monteith (Monteith, 1965) Penman (Penman, 1948) Priestly–Taylor (Priestly and Taylor, 1972) Hargreaves (Hargreaves and Samani, 1985) Baier–Robertson (Baier and Robertson, 1965)
Surface runoff	Green and Ampt method (Green and Ampt, 1911) for infiltration and rainfall excess rate Curve number (CN) (Mockus, 1969; USDA-NRCS, 2004) Options for daily CN adjustment: Variable daily CN nonlinear CN/SW with depth soil water weighting Variable daily CN nonlinear CN/SW no depth weighting Variable daily CN linear CN/SW no depth weighting Nonvarying CN–CN2 used for all storms Variable daily CN soil moisture index
Peak runoff rate	Modified rational formula (Williams, 1995) SCS TR55 Peak Rate estimate (USDA-SCS, 1986)
Subsurface flow	Storage routing and pipe flow equations
Crop growth	Potential growth Water use; nutrient uptake (N, N fixation, P) Growth constraints (water stress, temperature stress, nutrient stress, aeration stress) Biomass; root growth; crop yield Plant competition Winter dormancy
Available soil water capacity	Options for estimate field capacity and wilting point (if no good data): Estimated using the Rawls method (dynamic) Estimated using the Rawls method (static) Inputted using the Rawls method (dynamic) Estimated using nearest neighbor (dynamic) Estimated using nearest neighbor (static)
Soil erosion	
Water	Water erosion equations: Universal Soil Loss Equation (USLE) (Wischmeier and Smith, 1978) Onstad- Foster modification of the USLE (AOF) (Onstad and Foster, 1975) Revised USLE (RUSLE) (Renard et al., 1997) Modified USLE (MUSLE) (Williams, 1975) MUSLE variation 1 (MUST) (Williams, 1995) MUSLE variation 2 (MUSS) (Williams, 1995) MUSLE with input coefficients (MUSI) RUSLE2 (Foster, 2005)
	Slope length and steepness factor: RUSLE slope length and steepness factor (preferred for steep slopes >20%) MUSLE slope length and steepness factor

Continued.

Table 6–2. Continued.

Component	Method or principle
Wind	Wind erosion continuous simulation Surface roughness factor (Potter et al., 1990; Potter and Zobeck, 1990) Roughness factor Unsheltered distance factor (Stout, 1990) Wind erosion rate (Skidmore, 1986)
Nutrients	Nitrogen (organic N, denitrification, N cycling, nitrification, volatilization, rainfall N) Phosphorus (soluble P loss, P transport by sediment, mineralization, mineral P cycling)
Plant environmental control	Drainage, irrigation, fertilization, liming, pests, furrow diking

tion fields with a good grass cover, but manure erosion can be high. The models have manure management with automatic application from stockpile or lagoon, and a hail occurrence–damage component developed for use in crop insurance. The southern oscillation index is considered in the weather simulator for better rainfall definition during drought and for real time simulation.

Additional Features of APEX

As stated above, APEX extended EPIC's ability to simulate complex farm or watershed landscape processes and management practices. APEX can simulate detailed management practices related to farm animal productions, such as confined or partially confined area feeding, intensive rotational grazing, and cropland grazing after harvest. APEX has a spatial rainfall simulator for larger watersheds that provides daily rainfall distributed among subareas considering distance from the storm center and east–west or north–south gradients. The APEX routing component provides for physically based simulation of conservation practices like filter strips, terraces, and waterways. The routing mechanisms route water, sediment, nutrients, and pesticides across landscapes through channels, floodplains, and reservoirs to the watershed outlet. The routing feature of APEX enables the configuration of novel land management strategies. Soils are usually uniquely related to landscape position. Allowing the consideration of upper and lower landscape positions and representation of spatial locations of the various land uses and cultural practices can greatly improve modeling accuracy and allow alternative treatment such as best management practices in different landscape positions for maximum efficiency.

Input Data and Parameterization

The EPIC and APEX models require inputs of daily weather, soil attributes by layer, field management (e.g., planting crop, plant population, harvesting, tillage, fertilizer application rate and depth, pesticide application rate, irrigation), site information (e.g., latitude, longitude, elevation, CO_2 concentration in atmosphere, average concentration of N in rainfall), catchment characteristics (e.g., drainage area, average upland slope, average upland slope length), and channel geometry of routing reaches. The routing mechanism needs to be set up for an APEX run if a farm or watershed study involves multiple fields and subareas.

Climate Inputs

The weather variables required for EPIC and APEX simulations are daily precipitation, maximum and minimum air temperature, and solar radiation. Wind speed and relative humidity are also required if the Penman methods (Penman, 1948) are used to estimate potential evaporation. Wind speed is also needed when wind-induced erosion or dust emission and distribution are to be simulated. These weather variables can be all entered by the user or generated at run time. Both models also have options for using different combinations of both measured and generated weather variables. If any variables are user input (measured), then the daily precipitation data must be user input too.

For weather generation and other uses, monthly weather statistics are required. These include average monthly maximum and minimum air temperatures, monthly average standard deviations of daily maximum and minimum temperatures, average monthly precipitation, monthly standard deviation of daily precipitation, monthly skew coefficient for daily precipitation, monthly probabilities of wet day after dry day and wet day after wet day, average number days of rain per month, monthly maximum 0.5-h rainfall, average monthly solar radiation, and monthly average relative humidity. These variables are used to estimate missing daily data or generate daily weather variables (precipitation, maximum and minimum air temperatures, solar radiation, and relative humidity). Other uses include the monthly temperatures for calculating potential heat units (PHU, thermal time) for tree crops, the maximum 0.5-h rainfall for estimating peak runoff rate, and rainfall energy. The monthly wind weather statistics include average monthly wind speed and monthly wind percentage from 16 wind directions, which are critical if simulating wind erosion, as well as dust distribution and air quality from feedlots. These variables are used to generate daily wind speed and direction.

The monthly weather data can be prepared using historical daily weather data with a stand-alone weather generating program WXPM (Williams et al.,

2006). The models' database contains monthly weather statistics for about 1050 weather stations in the United States. If a long-term daily weather series is not available to generate the monthly weather statistics, the EPIC and APEX programs can reference the latitude and longitude of the simulation site to find the closest weather station in the database and use its monthly weather statistics.

Soils

The required soil data include soil albedo, hydrologic soil group, soil attributes by layer, including depth, bulk density, texture, organic carbon, and pH. The models have options to estimate missing soil attributes if soil inputs provide insufficient information. For example, the soil water contents at wilting point and field capacity can be estimated using either the Rawls method (Rawls and Brakensiek, 1985) or the nearest-neighbor method (Nemes et al., 2006).

Databases

The models have built-in databases for crops, tillage tools, fertilizers, and pesticides (Williams et al., 2006). The crop database contains information needed for EPIC and APEX to simulate the growth of a particular crop. The growth parameters in the plant growth database define plant growth under ideal conditions and quantify the impact of some stresses on plant growth. There are more than 100 plant species included in the crop database. When compiling the list of plants in the default database, an attempt was made to include the most economically important plants as well as those that are widely distributed in the landscape. However, the list is by no means exhaustive, and users may need to add plants to the list. When adding a new plant or land cover to the database, a review of existing literature should provide most of the parameter values needed to simulate plant growth. The tillage database contains information needed for EPIC and APEX to simulate the redistribution of nutrients and pesticides that occurs in a tillage operation and the surface roughness and bulk density changes caused by tillage operation. The common tillage management activities in agricultural land use are included. The fertilizer database summarizes nutrient fractions for various fertilizers and types of manures. The pesticide database contains properties of the active ingredients in pesticides. These databases are not comprehensive but have been designed so that users can easily append new information regarding crops, tillage tools, fertilizers, and pesticides.

APEX Routing Instruction

In APEX, there are only three variables, which need to be specified for each subarea to determine the routing structure of a watershed. In case of extreme areas

(subareas that receive no flow from another subarea) channel length (CHL) and routing reach channel length (RCHL) are identical (CHL = RCHL). For subareas that contain a routing reach (i.e., receive inflow from one or more subareas) CHL is greater or equal to RCHL. A negative watershed area (WSA) indicates that there is output from another subarea in storage that must be added to the subarea output. Therefore, it is possible to manually construct APEX's routing structure (Williams et al., 2006). For a larger watershed or a complex landscape, geographic information system (GIS) tools are available to build APEX inputs. For example, the ArcAPEX tool is designed to use ArcGIS 9.x GIS platform to prepare APEX inputs (Tuppad et al., 2009). The GIS layer of ???DEM is required to define the catchment outer boundary and also to delineate the stream networks within the catchment. The subarea in APEX is defined using the resulting stream networks from the DEM delineation such that each subarea is draining the area upstream of a certain location in the stream networks. User-defined streams and subarea boundaries, which are closely associated with specific agricultural field boundaries, can be used in place of the DEM-based delineation in ArcAPEX. User defined subareas could be delineated from readily available aerial imagery, e.g., United States Department of Agriculture-National Agriculture Imagery Program (USDA, 2009) or other remote sensing datasets.

Calibration and Validation

Comprehensive models like EPIC and APEX contain a small number of parameters that are subject to calibration by the user and a large number of parameters that are contained in the algorithms and are inaccessible to the users. For that reason, the calibration process, being either manual or automated, requires careful inspection of outputs, good judgment, and above all a disclosure of the parameters adjusted to obtain good agreement between observed and simulated data. Model calibration is to adjust influential model parameters or inputs within their reasonable ranges so that the modeling results (see major outputs in Table 6–3) are comparable with observed data. Normally some variables fit measured data better than others because they are relatively independent. For example, sediment yield, nutrients, and pesticides depend on runoff so errors in simulating runoff can propagate in those variables. Corrective measures can be taken in the calibration by identifying the actual source of the disagreement.

The first step before any calibration is to examine the EPIC or APEX output file "*.out" from a preliminary model run. This output file provides comprehensive information about the simulation (inputs; daily, monthly, or annual simulation results; and summary). Therefore, it is the most important file for iden-

Table 6–3. The EPIC and APEX models' input files likely to be updated during calibration and major model output files. See user manuals in Williams et al. (1990, 2006) for detail.

File		Short description
		Input
Model control	*cont.dat	Includes a variety of data parameters, e.g., submodel options can be indicated in this file
Model parameter	parm*.dat	Contains definitions of s-curve used to describe the behavior of many processes and PARM1-PARM83 mainly for defining equation coefficients
Soil attributes	*.sol	Lists soil attributes, such as hydrologic soil group, years of cultivation before simulation, and other soil attributes by layer
Field management	*.opc or *.ops	Lists operation schedule by date, such as planting (crop ID, potential heat units, land use number or CN2, plant population), harvesting, tillage, fertilizer application (fertilizer ID, application rate, depth), pesticide application (pesticide ID, application rate)
Subarea information	*.sub	Contains subarea information, such as soils and operation schedules used for each subarea, catchment characteristics, and channel geometry of routing reaches
Crop database	*crop.dat	Lists plant species and plant growth parameters that define plant growth under ideal conditions and quantify the impact of some stresses on plant growth
		Output
Detailed simulator	*. out	Provides comprehensive information about the simulation (inputs/options used; daily, monthly, or annual simulation results; and summary)
Annual crop yield	*.acy	Annual crop grain, fiber, and forage yield, biomass; stress days of drought, N, P, temperature, and water; nutrients in the root zone for each subarea
Annual subarea	*.asa	Annual evapotranspiration, surface runoff, subsurface flow, percolation, drainage flow, C loss with sediment, erosion–sedimentation, soluble N and P losses, organic N and P losses for each subarea
Annual watershed	*.aws	Annual upland and watershed flow, sediment yield, C loss with sediment, soluble N and P losses, organic N and P losses
Daily subarea pesticide	*.dps	Pesticide name; amount applied; losses from runoff, percolation (root zone), subsurface flow, sediment, and drainage flow; pesticide degradation from foliage and soil
Annual soil organic C and nitrogen	*.acn	Bulk density; C pools of biomass, slow humus, passive humus, and total C pool; structural litter N; metabolic litter N; N pools of biomass, slow humus, passive humus, and total N pool for each soil layer and a total across layers
Daily subarea hydrology	*.dhy	Daily surface runoff, subsurface flow, quick return flow, water yield, peak flow rate, root zone soil water, groundwater storage
Monthly watershed file	*.mws	Watershed flow, soil loss from erosion, soluble N and P losses, organic N and P losses
Reach output	*.rch	Area of subarea or reach outlet; daily, monthly or annual reach in and out variables of flow, sediment, soluble N and P loads; organic N and P loads

tifying possible causes for poor simulation results. A basic check is to see if the total watershed area obtained by adding all subareas during APEX simulation is correct. Incorrect total watershed area indicates mishandling of the routing instructions in the input files. This may occur if the routing instructions are prepared manually. Water balance for the entire watershed should be checked. Average annual surface runoff, water yield, sediment yields and nutrient yields should be checked. If they are not reasonable, then check individual subarea contributions to narrow down possible causes. The user needs to decide in this case what a reasonable outcome is. It is extremely important to use actual information to judge model outcomes, and to avoid imposing our expected outcome by adjusting calibration parameters ("drawing the output").

After making sure that the fertilizer/manure rates, water and nutrient dynamics are reasonable, limited adjustment can be performed based on the comparison between predicted and observed data. After the preliminary investigation, the influential model parameters should be identified based on sensitivity analysis, experience, or literature. Useful guides can be found in Wang et al. (2005, 2006c) and Yin et al. (2009). Sensitivities are dynamic in both temporal and spatial dimensions; therefore, a site-specific sensitivity analysis is encouraged.

Automatic calibration procedures with Monte Carlo simulations and multi-objective functions have been successfully used in EPIC (Wang et al., 2005) and APEX (Yin et al., 2009) calibration. Zhang et al. (2010) developed a multi-objective optimization algorithm coupled with EPIC for exploring the trade-offs between biofuel energy production and unintended ecosystem-service responses. The algorithm could be adopted for parameter optimization where measured data are available. However, model users should use good judgment when analyzing outputs from automatic and manual calibration considering errors in measured variables and inputs. The widespread argument that a model may be tuned for a right answer but with a wrong reason, is valid. Therefore, it is always a good practice to look into various outputs during interactive manual calibration.

The calibration should first consider a correct water balance, including all components, and a correct simulation of erosion, crop growth, and nutrient cycling. These processes are intimately intertwined, and the calibration requires interpretation. A description of a standard procedure for calibrating and testing EPIC and APEX is provided in the following sections. Given the large number of processes simulated, the procedure is generic and needs to be tailored to the specific simulation under consideration. Table 6–3 lists the model input files where some inputs and parameters are likely to be adjusted during calibration.

Calibration of Hydrologic Balance

The hydrologic balance component encompasses the key processes that occur in the hydrologic cycle. Initially, rainfall may be intercepted by plant canopy or fall directly on the soil surface. Incoming precipitation, snowmelt, or irrigation is partitioned between surface runoff and infiltration. Infiltrated water stored in the soil profile is subject to subsurface flow and evapotranspiration (ET). The vertical subsurface flow may leave the root zone and enter the groundwater storage, where it is partitioned into deep percolation from the system and return flow to channel. Horizontal flow within the root zone is partitioned into lateral and quick return flow. The algorithms to represent these processes have a mix of mechanistic equations and empirically derived equations and associated parameters to simulate mass transport.

Potential Evapotranspiration

The five PET methods included in the models (Table 6–2) vary in the amount of required weather inputs. The Hargreaves method (Hargreaves and Samani, 1985) provides a simple and relatively robust method for estimating PET that only requires daily maximum and minimum temperature. The Penman–Monteith method is generally considered the most accurate but requires solar radiation, air temperature, relative humidity, and wind speed, which are not always available from measurements. The prediction of PET by Penman–Monteith can be adjusted through the stomatal conductance coefficient PARM1 (Table 6–4). The Priestley–Taylor method is a simplification of the Penman–Monteith method. It requires solar radiation and air temperature. Like the Hargreaves method the Baier–Robertson method requires minimal data; however, it was developed in Canada and is provided mainly as an alternative for use in cold climates. Because the Hargreaves PET equation requires minimal inputs and is commonly used in EPIC–APEX applications, only this method is described in this chapter.

$$\mathrm{PET} = \mathrm{PARM23}\left(\frac{\mathrm{RAMX}}{2.5 - 0.0022T_{\mathrm{x}}}\right)\left(T_{\mathrm{a}} + 17.8\right)\left(T_{\mathrm{x}} - T_{\mathrm{n}}\right)^{\mathrm{PARM34}} \quad [1]$$

where PARM23 (range 0.0023–0.0032) and PARM34 (range 0.5–0.6) are empirical coefficients, RAMX is the clear sky daily radiation at the site elevation (MJ m^{-2} d^{-1}), T_a is the mean daily air temperature (°C), T_x and T_n are the daily maximum and minimum air temperatures (°C), Increasing PARM34 increases PET and decreases soil water and crop yield because transpiration and growth are decoupled in these models when using the radiation use efficiency to simulate growth.

Surface Runoff

The USDA-NRCS runoff curve number (CN) as modified for continuous simulation is a reliable method for estimating surface runoff given daily precipitation. The models vary CN continuously as soil water changes through the retention parameter, *S*. The models have four different methods of linking *S* (CN) and soil water plus a constant CN option (Table 6–2). Generally the soil moisture index (Option 5, see Table 6–2) is relatively robust and reliable. Kannan et al. (2008) has demonstrated that this method can produce realistic runoff estimates over a wide range of soils. Because this method is commonly used in EPIC–APEX applications, all the other methods also have no provision for calibration. Therefore, only this method in which two parameters (PARM42 and PARM44) can be calibrated, is described in this chapter. This method calculates the daily retention parameter *S* accounting for plant evapotranspiration using the equation:

$$S = S_{\text{prev}} + \text{PET}\exp\left(-\text{PARM42}\frac{S_{\text{prev}}}{S_{\max}}\right) - P_{\text{prev}} + Q_{\text{prev}} + Q_{\text{return}} + \text{SSF} + \text{PRK}; \qquad S < \text{PARM44}S_{\max} \tag{2}$$

where *S* is the retention parameter for a given day (mm), S_{prev} is the retention parameter on the previous day (mm), PET is the potential evapotranspiration for the day (mm d^{-1}), PARM42 is the CN index coefficient, PARM44 is the upper limit of CN retention parameter, $S_{\max}$ is the maximum value the retention parameter can achieve (mm), which is associated with CN_1 (the curve number for moisture condition 1 or dry condition), P_{prev} is the rainfall reaching the ground surface after plant interception on the previous day (mm), and Q_{prev} is the runoff on the previous day (mm), Q_{return} is quick return flow on the previous day (mm), SSF is the lateral subsurface flow on the previous day (mm), and PRK is percolation on the previous day (mm).

Considerable runoff can be infiltrated into floodplains depending on travel time and floodplain saturated conductivity. Flow through a routing reach can be increased by return flow from groundwater and quick return flow. Groundwater storage must exceed a threshold value before return flow occurs. Therefore, the floodplain saturated conductivity, groundwater storage threshold, maximum groundwater storage, groundwater residence time, and ratio of return flow to return flow plus deep percolation can be used to adjust runoff loss or gain during routing (Table 6–4).

Table 6–4. Calibration of major EPIC and APEX outputs.

Output	Influential process	Influential input or parameter	Range	Input file
Runoff	PET equations (indicating option in *cont.dat)	If Hargreaves, PARM34 If Penman-Monteith, PARM1	0.5–0.6 1.0–2.0	parm*.dat
	Crop growing season	Potential heat units (PHU) planting and harvest dates If automatic heat unit schedule (HUSC)	1.0–1.2	*.opc or *.ops
	Runoff equation– curve number equations (indicating option in *cont.dat)	CN2 or land use number (LUN)		*.opc or *.ops
		If variable daily soil moisture index, PARM42	0.5–2.5	parm*.dat
	Irrigation (if used)	Irrigation runoff ratio (EFI)		*.sub or *.opc
	Routing	Floodplain saturated conductivity (FPSC) Maximum groundwater storage (GWSO) Groundwater residence time (RFTO) Return flow/(return flow + deep percolation) (RFPO)		*cont.dat (values overridden by values > 0. in *.sol)
		Groundwater storage threshold: PARM40	0.001–1.0	parm*.dat
Crop yield	Crop growing season	See above		
	Growth constraints (check stress days in *.out) Root growth stress Water stress: excessive PET or runoff low plant available water Nutrient stress: low mineralization rates inadequate fertilization excessive leaching of N	High bulk density (BD) or aluminum saturation caused by low pH < 5 (Applying deep tillage and lime to lower BD and pH, respectively.)		*.sol (if correcting inputs)
		See above for related parameters Difference between field capacity (FC) and wilting point (WP)		*.sol (if correcting inputs) *cont.dat (estimation option)
		Organic N, P, and C Years of cultivation at simulation (RTN1) Fertilization rates; or if auto fertilization: Auto-fertilization application rate (FNP4) Max. annual N application for a crop (FMX) Soil plant-available water (FC- WP) saturated conductivity	C/N near 10 N/P near 8	*.sol *.sub

Continued.

Table 6–4. Continued.

Output	Influential process	Influential input or parameter	Range	Input file
Crop yield, continued				
		Plant population		*.opc or *.ops
		Winter dormancy, PARM6	0–1	*parm.dat
		Maximum potential leaf area index (DMLA) Harvest index (HI) Biomass-energy ratio (WA)		cropcom.dat
Erosion–sedimentation	Erosion/sedimentation equations (indicating option in *cont.dat)	RUSLE C factor coefficient: PARM46 PARM47	0.5–1.5 0.01–3.0	parm*.dat
		Erosion control factor (PEC)	0.0–1.0	*.sub
		Slope length and steepness factor equations		*cont.dat
	Routing	Sediment routing exponent, PARM18 Sediment concentration at 1.0 m s^{-1} velocity, PARM19	1.0–1.5 0.005–0.25	parm*.dat
		Channel and floodplain geometry Normally unknown, if let the model estimate, then input: channel capacity flow rate in mm h^{-1} (QG)	about 2 yr frequency flow rate	*cont.dat
		Channel cover factor (RCHC) Soil erodibility factor (RCHK)	0.001–1.0 0.001–1.0	*.sub
Nutrient	Nitrogen cycling Carbon cycling Phosphorus cycling	Biological mixing efficiency, PARM29 Soluble P runoff coefficient, PARM8 P upward movement by evaporation coefficient, PARM59 Nitrate leaching ratio, PARM14 Denitrification soil_water threshold, PARM35 N fixation coefficient, PARM7 Volatilization/nitrification partitioning coefficient, PARM72 Microbial decay rate coefficient, PARM70 Fraction of humus in passive pool, FHP	0.1–0.5 10–20 1–30 0.1–1.0 0.9–1.1 0.0–1.0 0.05– 0.5 0.05– 1.5 0.3– 0.9	parm*.dat *.sol

Calibration of Crop Grain Yield

The models are capable of simulating growth for both annual and perennial crops. Phenological development of a crop is based on daily heat unit accumulation. Annual crops grow from planting to harvest or until the accumulated heat units equal the potential heat units for the crop. Perennial crops maintain their root systems throughout the year, although they may become dormant after frost. The model is also capable of simulating mixed stands of up to 10 crops or other plants in a competitive environment.

If crop yields are available, they can be calibrated together with runoff calibration or after runoff calibration. Realistic PET and proper crop growing season are important to both runoff and crop yield. Unreasonable ET are usually associated with a crop not emerging or stressed due to lack of water and/or nutrients. Excessive PET or runoff estimates can reduce crop yields because it can increase crop water stress. Plant-available water is another important limitation that causes water stress. Erroneous estimates of plant-available water occur when field capacity (FC) or wilting point (WP) are incorrect. Soil water storage is particularly important in dry climates. Soil depth may also curtail crop yield. Plant population and heat units from planting to harvest impact crop grain yield as well. Crop parameters contained in the crop database may need to be adjusted, including biomass energy ratio (WA), base temperature (T_b), harvest index (HI), and maximum leaf area index (LAI_{mx}) (Table 6–1).

Root growth stresses due to bulk density (BD) or aluminum toxicity (ALSAT) can reduce crop yields greatly. Bulk density can be lowered by deep tillage or the addition of crop residue. Lime may be applied automatically by the models to increase pH and reduce aluminum toxicity stress.

Nitrogen and P stress may be caused by low mineralization rates, inadequate fertilizer, gaseous losses, or excessive leaching of N (Table 6–4). The soil organic C/N should be near 10, and N/P should be near 8. The mineralization rate can be increased by decreasing the number of years of cultivation at the beginning of simulation (RTN1). High N leaching (PRKN) relative to annual N fertilizer results from high percolation (PRK), which may result from low ET or runoff, low soil plant-available water storage (FC–WP), or high saturated conductivity values.

Calibration of Erosion–Sedimentation

Runoff must be realistic before calibrating erosion–sedimentation. Proper cover and residue are critical; therefore, the crop growth must also be realistic. For watershed analyses, the water erosion equations of MUSLE, MUSS, and MUST are recommended. The three equations usually give similar results and are appropriate for estimating sediment yield from small watersheds up to about 250

km^2. The universal soil loss equation (USLE) is an erosion equation that is useful in studies assessing the effect of erosion on productivity. These equations are in the following form:

$$Y = X \times \mathrm{EK} \times \mathrm{CVF} \times \mathrm{PEC} \times \mathrm{SL} \times \mathrm{ROKF} \quad [3]$$

USLE and RUSLE???

$$X = \mathrm{EI} \quad [4]$$

MUSLE:

$$X = 1.586\left(Qq_{\mathrm{p}}\right)^{0.56} \mathrm{WSA}^{0.12} \quad [5]$$

MUST:

$$X = 2.5\left(Qq_{\mathrm{p}}\right)^{0.5} \quad [6]$$

MUSS:

$$X = 0.79\left(Qq_{\mathrm{p}}\right)^{0.65} \mathrm{WSA}^{0.009} \quad [7]$$

where Y is the sediment yield (Mg ha^{-1}), X is the rainfall and runoff factor, EI is rainfall energy factor for USLE, EK is the soil erodibility factor, CVF is the crop management factor, PEC is the erosion control practice factor, SL is the slope length and steepness factor, ROKF is the coarse fragment factor, Q is the runoff volume (mm), q_{p} is the peak runoff rate (mm h^{-1}), and WSA is the watershed area (ha).

Because the plant cover varies during the growth cycle, the CVF factor is internally updated for all days when runoff occurs. It is calculated as a function of crop residue, crop height, standing live biomass of the crop, and soil surface random roughness using the following equations:

$$\mathrm{CVF} = \exp\left[-0.026(\mathrm{RRUF} - 6.1)\right] \mathrm{FRSD}\ \mathrm{FBIO} \quad [8]$$

where RRUF is the soil surface random roughness (mm), FRSD is the crop residue factor, and FBIO is the growing biomass factor. The FRSD and FBIO factors are calculated using the following equations:

$$\mathrm{FRSD} = \exp(-\mathrm{PARM46}\ \mathrm{CVRS}) \quad [9]$$

$$\mathrm{FBIO} = 1.0 - \exp(-\mathrm{PARM47}\ \mathrm{CPHT}) \frac{\mathrm{STL}}{\mathrm{STL} + \exp(1.175 - 1.748\mathrm{STL})} \quad [10]$$

where CVRS is the aboveground crop residue (Mg ha^{-1}), PARM46 and PARM47 are coefficients in the exponential functions, CPHT is the crop height (m), and STL is the standing live biomass of the crop (Mg ha^{-1}).

Increasing PARM46 decreases FRSD and CVF, which leads to lower sediment yield estimates. The effect of crop height on the CVF factor is governed by the exponential coefficient PARM47. Increasing PARM47 diminishes the effect of ground cover (represented by STL in Eq. [10]), increases FBIO and CVF, which increases sediment yield estimates. The two parameters can be adjusted for sediment yield calibration.

The channel vegetative cover factor (RCHC) ranges from near 0.0 for a channel with extremely good cover, like a grass waterway, to about 0.8 for a channel essentially bare of vegetation. The soil erodibility factor (RCHK) ranges from near 0.0 for channels with extremely stable side slopes and bottom to about 0.5 for shallow channels in cropped fields. PARM19 is the sediment concentration at a flow velocity of 1.0 m s^{-1}. The concentration (t m^{-3}) should normally range from 0.01 to 0.05. High values cause overestimation of sediment transport. These parameters and the sediment routing exponent (PARM18) may be adjusted for sediment yield calibration.

Calibration of Nutrients

The first item to check is the application rate of fertilizer/manure; then check the individual components (denitrification, volatilization, leaching, runoff, crop uptake, and N fixation). Parameters that may be adjusted to improve results include PARM29 (biological mixing efficiency), PARM14 (the ratio on N concentration in the runoff to that of the percolate), PARM72 (partitions between volatilization and nitrification), PARM7 (N fixation coefficient ranges from 0.0 for the highest estimate to 1.0 for the lowest, PARM8 (P linear adsorption coefficient—inversely related to soluble P in the runoff), PARM70 (microbial decay rate coefficient), and FHP (fraction of humus in passive pool) (Table 6–4).

Application

The APEX model is used in an ongoing Special Emphasis Conservation Effects Assessment Project (CEAP) watershed study for the North Bosque watershed (322,700 ha) in central Texas (Plate 6–1; see color insert section). In 2000, the North Bosque River was listed as an impaired water body in the Texas Water Quality Inventory for concerns of elevated levels of bacteria, chlorophyll a, and nutrients entering the segment from tributary watersheds. Upper North Bosque was also placed on the 303(d) list for elevated levels of sediment, nitrogen, phosphorus, chloride, sulfate, and chlorophyll a. These impairments have mainly been associ-

ated with the dairy industry in the northern part of the watershed. The objective of the CEAP watershed study is to identify conservation practices to reduce the phosphorous loading levels flowing from the North Bosque watershed into Lake Waco by testing alternative practices and evaluating the timing, location and suites of practices that trap, filter, or otherwise mitigate phosphorus loss. APEX has been calibrated at Hico (Plate 6–1) on the basis of available historical data. The following sections mainly focus on the calibration and validation of the APEX model for monthly flow, sediment, nitrogen, phosphorous stream loadings from the North Bosque watershed above Hico.

Watershed Description and Input Data

The North Bosque originates in Erath County and flowing southeast through Bosque and McLennan Counties where it is impounded to create Lake Waco. The lake provides 75% of the water supply for the city of Waco, TX. In the early 2000s the study area has the largest concentration of dairy animals in the state of Texas. There are an estimated 9300 ha of manure application fields. In the CEAP watershed study, it was assumed there are 61 permitted dairy farms (Plate 6–2) with 39,825 confined dairy cows. Some estimates indicate the number of dairy cows may be around 40,000. However, no effort was made to actually quantify the number of cows in each of the individual dairies; because of variations by season and by years depending on many external factors, the number cannot be verified. We used a percentage of the permitted number of cows to distribute the cows among the individual dairy farms.

The watershed area of the North Bosque watershed above Hico is 92,650 ha. Hico is located in the very center of the North Bosque watershed (Plate 6–1). This subwatershed contains 41 of the 61 dairies in the North Bosque (Plate 6–2). Waste application fields identified in the Texas Institute of Applied Environmental Research (TIAER) and the Texas Commission on Environmental Quality (TCEQ) database were used to locate the waste application fields within the subwatersheds. We did not have an accurate account of which waste application fields were used by which dairy, so we assigned the fields to the nearest dairy. This assumption, however, is likely to cause some misrepresentation of the actual conditions. Unless we had specific information to assign land use, all waste application fields were assumed to have coastal bermudagrass [*Cynodon dactylon* (L.) Pers.] as the land cover. These assumptions, however, unavoidably contribute to the uncertainty of model inputs.

Daily weather data were obtained from historical weather records of the National Oceanic and Atmospheric Administration (NOAA) National Weather Service. Ten stations (three above Hico) were chosen that had both temperature and precipita-

tion records from 1965 to 2004 for the CEAP watershed study. The long-term average annual precipitation for this watershed is 842 mm. The average monthly maximum temperature is 25°C, and the average monthly minimum temperature is 11°C. The 40-yr period was used for various scenario analyses in the CEAP study. However, observed data of stream flow, sediment, and nutrient losses are only available at the Hico monitoring station for 1993 through July 1998. In this study, the simulated monthly data were compared with the available monthly measured data at Hico.

A new land use–land cover inventory was taken for the CEAP watershed study. Unsupervised classification was used to stratify 300 points into 25 categories each using 2002 Thematic Mapper satellite data. Of these 300 points, a total of 260 points could be reached and classified by ground observation. These points were used to classify the entire watershed for land use–land cover as of 2002. The land use was 63.5% native pastures–Range, 14.2% improved pasture(assumed to be improved Bermuda grass), 10.4% misquote–cedar mix, 6.4% deciduous trees (including wetlands), 4.3% cropland, 1% Urban, and 0.2% Dairies and milk sheds. Plate 6–3 shows the land use–land cover and the Soil Survey Geographic (SSURGO) soils data for the watershed above Hico.

The 10-m elevation data from Texas Natural Resource and Information Systems (TNRIS) were used to develop the detailed watershed data. The watershed attributes required for APEX include delineation of the subarea boundaries, estimation of the drainage system, slope and slope length, channel lengths, and subbasin dimensions (length and width). The philosophy of the study design was to divide the Bosque basin into small enough subunits so as to let one subbasin represent one field or pasture. After some experimentation and generation of subbasins using GIS tools and viewing these with the land use and the National Agriculture Imagery Program (NAIP) data, the North Bosque watershed was divided into a total of 7544 subbasins. These subbasins were again divided into upper and lower landscape positions to make a total of 15,088 subbasins, among which 3475 subareas are above Hico. Plate 6–4 shows the upper and lower landscape positions within one 12-digit HUC??? in the study watershed. This separates the lower position areas in the landscape to allow alternative treatment placements. The separation of the landscape into two positions was accomplished using a GIS algorithm that performed several functions:

1. The algorithm calculated the position of each 10 by 10 m cell by assigning a percentage distance between the stream line and the ridge line of each of the subbasin.
2. It then calculated the change in slope for each of the cells.
3. The next step was to identify the cells that had the most rapid change in slope found in the lower 40% of the subbasin.

4. The algorithm finally assigned the cells below the maximum slope change to the lower landscape position and above that line to the upper landscape position. The set of rules required that all upper landscape positions flow into or through a lower landscape position before entering the stream channel. The lower landscape position was assumed to have a drainage channel receiving water from the immediately adjacent upper landscape position.

Cropping systems were chosen that represented crops and management practices in the area.

Coastal bermudagrass was used for all waste application areas. This constituted the planning of the coastal bermudagrass and harvesting of hay from the waste application area four times a year. The pasture and range areas identified from the land cover classifications were assigned various combinations of grasses and shrubs, including Mesquite trees [*Prosopis chilensis* (Molina) Stuntz], little bluestem grass (*Dichanthium* Willem.), and side oats grama grass [*Bouteloua curtipendula* (Michx.) Torr.]. The grass on this land cover was harvested three times a year to simulate livestock grazing. Heavy woodlands were simulated by a cover of deciduous trees with no harvesting of biomass was removed from these areas. Improved pasture was simulated either as coastal bermudagrass or buffalograss [*Bouteloua dactyloides* (Nutt.) J.T. Columbus], with commercial fertilizer used. A simple mix of primary crops was chosen for croplands. Three crops—corn, sorghum [*Sorhum bicolor* (L.) Moench], and wheat—were placed into three rotations for the study. These rotations were (i) continuous corn, (ii) corn–sorghum, and (iii) wheat–sorghum. The automatic fertilizer (nitrogen application) option was used to provide the nutrient for these rotations.

There are two primary types of reservoirs found in the study area. First are those constructed by the USDA-NRCS as flood prevention reservoirs. Most of these reservoirs were constructed in the 1950s and 1960s. The Bosque watershed has 41 such reservoirs located in the entire basin above Lake Waco. The second type of reservoir includes private, water supply reservoirs and those built by other organizations. We identified 34 of these reservoirs to be included for the CEAP watershed study. A total of 41 reservoirs were within the watershed above Hico. The data for both types dams and reservoirs are found in the national inventory of dams (http://geo.usace.army.mil/pgis/f?p=397:12:1816408540398851 [verified 1 Apr. 2011]). This inventory provides sufficient information to allow us to parameterize each of these individual dams found in the watershed.

The ArcGIS toolbox and the APEX interface, WinAPEX (Magre et al., 2006), were used to process and manage the databases and model runs. WinAPEX has built-in algorithms that extracts the data from a MS Access database and formats the properties into the respective files needed by APEX.

APEX Calibration and Validation

The monthly record of stream flow, sediment yield, mineral N, mineral P, organic N, and organic P available at the Hico monitoring station were used to calibrate and validate the APEX model. About one-half of the observed data (July 1995–July 1998, i.e., 3 yr) were chosen for calibration because this period includes the biggest and smallest rainfall events. The remaining data (January 1993–June 1995) were used for validation. The model options chosen for this study were the USDA-NRCS curve number (CN) method for runoff estimation, variable daily CN soil moisture index method to estimate daily CN, modified rational equation to calculate peak flow, the Hargreaves method to calculate potential evapotranspiration, a variation of the modified Universal Soil Loss Equation—the MUST equation (Williams, 1995)—to calculate erosion/sedimentation (Table 6–2), and the GLEAMS enrichment ratio method (Leonard et al., 1987) for P transport and transformation.

The parameters adjusted for flow calibration included PARM42 (1.0), PARM40 (0.1), and RFPO (0.5). The parameters adjusted for sediment yield included PARM46 (0.62), PARM47 (1.0), PARM18 (1.0), and PARM19 (0.007). The calibration of N and P losses were conducted by adjusting PARM8 (10), PARM14 (0.09), PARM29 (0.2), PARM35 (1.1), and PARM59 (10). These parameters were described in Table 6–4. The values within the parentheses denote the actual calibrated values. Statistical measures including mean, standard deviation, R^2, Nash–Sutcliffe efficiency (NSE) (Nash and Sutcliffe, 1970), percentage bias (PBIAS), and the ratio of the root mean square error to the standard deviation of observed data (RSR) were used to evaluate the model performance based on criteria suggested by Moriasi et al. (2007).

Results and Discussion

The simulated monthly stream flow, sediment yield, and nutrient losses compared well with observed values for the calibration period, as evidenced by the values of NSE (0.69–0.85), R^2 (0.70–0.92), PBIAS (−18–8%), and RSR (0.39–0.56) in Table 6–5. Based on the statistical criteria (established based on values of NSE, RSR, and PBIAS) for evaluating water quality model performance proposed by Moriasi et al. (2007), the model performance is good for monthly flow, sediment, and nutrient losses during the calibration period (July 1995–July 1998). During the validation period (January 1993–June 1995) the model performance was good for monthly flow, mineral P, and Total P, for which NSE values ranged from 0.74 to 0.78 and RSR values from 0.49 to 0.56 and satisfactory for monthly sediment, organic P, and total N, for which NSE values ranged from 0.54 to 0.62 and RSR values from 0.68 to 0.70. However, the model performance was unsatisfactory for organic N and mineral N losses during the validation period. This could partly

Table 6–5. Summary statistics of monthly calibration and validation results for flow ($m^3\ s^{-1}$), sediment (Mg ha^{-1}), and nutrient (kg ha^{-1}) at Hico monitoring station.†

		Observed		Simulated		NSE	R^2	PBIAS	RSR
		Mean	SD	Mean	SD				
								%	
Calibration (July 1995– July 1998) *N* = 36	Flow	5.07	6.05	5.10	7.28	0.85	0.92	0.7	0.39
	Sediment	0.05	0.12	0.06	0.08	0.81	0.84	8.0	0.42
	Mineral P	0.03	0.05	0.03	0.04	0.69	0.70	−13.6	0.56
	Organic P	0.04	0.07	0.04	0.05	0.77	0.79	−5.1	0.47
	Mineral N	0.10	0.12	0.10	0.13	0.70	0.73	5.5	0.55
	Organic N	0.24	0.36	0.20	0.26	0.77	0.83	−18.1	0.49
	Total P	0.08	0.12	0.07	0.09	0.77	0.79	−8.8	0.47
	Total N	0.34	0.47	0.30	0.38	0.79	0.82	−11.2	0.46
Validation (January 1993– June 1995) *N* = 31	Flow	3.54	4.17	3.54	4.20	0.74	0.76	0.0	0.54
	Sediment	0.04	0.07	0.04	0.05	0.62	0.64	−10.6	0.70
	Mineral P	0.02	0.03	0.02	0.04	0.78	0.84	28.6	0.49
	Organic P	0.03	0.04	0.04	0.04	0.61	0.62	5.1	0.70
	Mineral N	0.08	0.09	0.09	0.11	0.17	0.52	8.3	0.94
	Organic N	0.20	0.28	0.16	0.21	0.46	0.48	−21.7	0.73
	Total P	0.05	0.07	0.06	0.08	0.74	0.76	13.4	0.56
	Total N	0.28	0.35	0.24	0.30	0.54	0.57	−13.2	0.68

† NSE, Nash–Sutcliffe efficiency; PBIAS, percentage bias; RSR, ratio of the root mean square error to the standard deviation of observed data.

be attributed to the uncertainty in model input data in the N fertilizer application with the assumption that commercial fertilizer was used for improved pasture and the automatic N fertilizer option in APEX was used for cropland, and that no commercial fertilizer was applied for the remaining land uses. Nutrient loads were relatively lower during the validation period than in the calibration period. With recognized uncertainty in measured data (Harmel et al., 2006; Harmel and Smith, 2007), low loads might contribute to the relatively weaker comparison during the validation period because low loads may be difficult to detect therefore with higher uncertainty. Harmel et al. (2006) indicated that model results within 10 to 31% of the measured values are within the average uncertainty range of water quality data measured with a typical quality assurance–quality control effort. Moriasi et al. (2007) also proposed that the values of PBIAS (percentage error or bias) for simulated average monthly N or P within ±25 to ±40% of observed values are acceptable. The PBIAS values in this study were within ±20%, except for mineral P (28.6%) and organic N (–21.7%) (Table 6–5).

Simulated monthly flow, sediment yield, total N, and total P at the Hico monitoring station followed, in general, the patterns of observed values during both

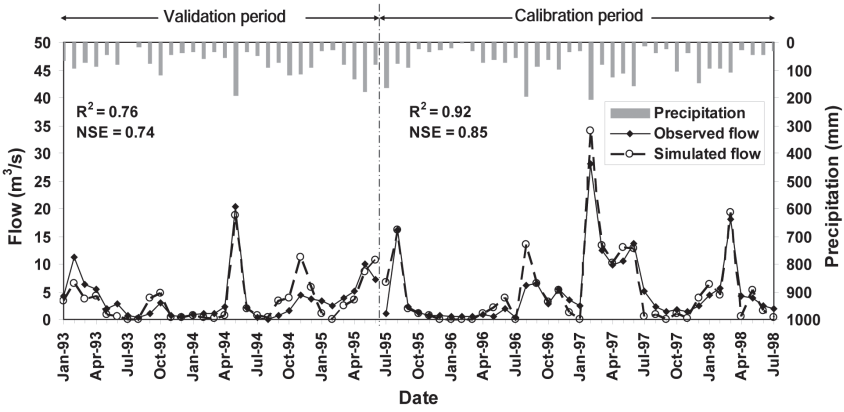


Fig. 6–2. Observed and simulated flow for the calibration and the validation periods at the Hico monitoring station.

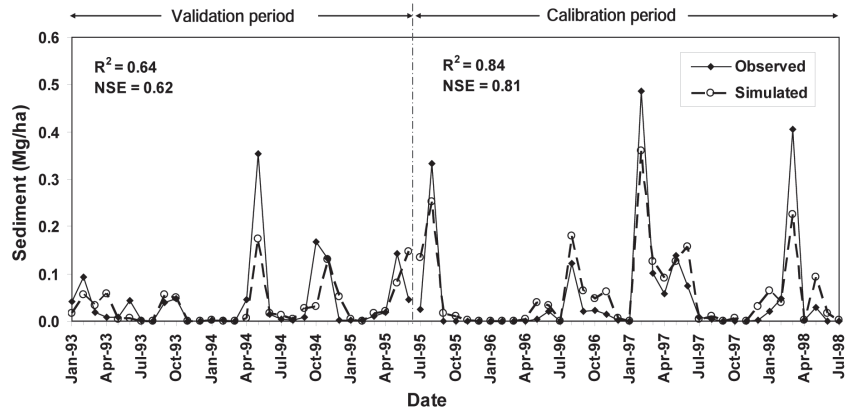


Fig. 6–3. Observed and simulated sediment yield for the calibration and the validation periods at the Hico monitoring station.

the calibration and the validation periods (Fig. 6–2 through 6–4). Total N and total P losses were clearly underestimated in April 1993 and February 1997. However, APEX overestimated the total N and total P losses in September 1993 and August 1996 (Fig. 6–4 and 6–5). One possible reason for the discrepancies between the observed and simulated monthly nutrient losses could be that the actual number of cows in each of the individual dairies, as well as seasonal or annual changes to their counts, were unknown. The actual waste application field for each of the individual dairies was also unknown. Further calibration by adjusting the number of cows or seasonal changes in each dairy or adjusting the assignment of waste application fields for each dairy may improve the results. Plate 6–5 shows the total P loss from each subarea in the study watershed. The high total P losses

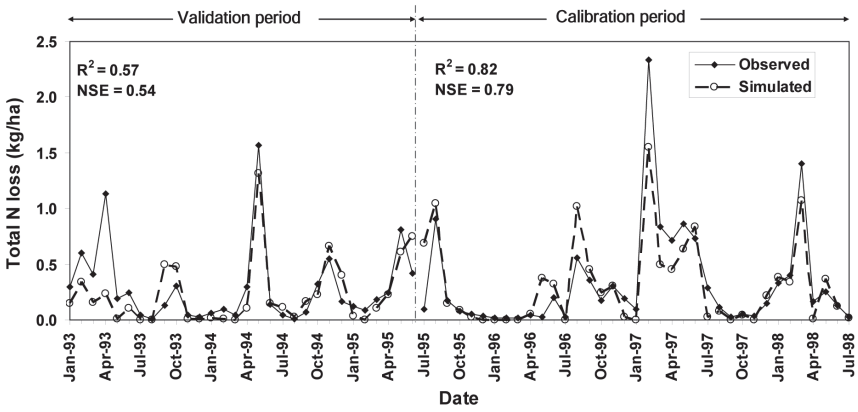


Fig. 6–4. Observed and simulated total N loss for the calibration and the validation periods at the Hico monitoring station.

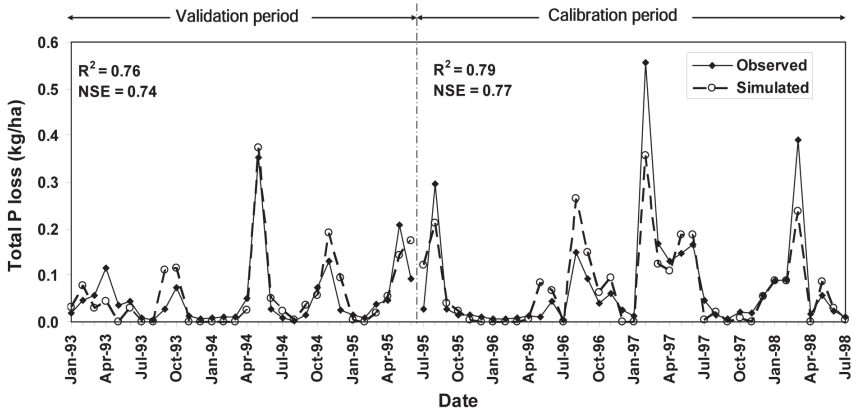


Fig. 6–5. Observed and simulated total P loss for the calibration and the validation periods at the Hico monitoring station.

were from the waste application fields, especially where the waste application areas were inadequate to support the waste application from nearby dairies.

Conclusions

Both EPIC and APEX are comprehensive terrestrial ecosystem models capable of simulating the basic biological, chemical, hydrologic, and meteorological processes of farming systems and their interactions. The models dynamically represent multiple soil properties including depth and bulk density for each soil layer during simulations. They remove eroded soil, attached organic nutrients, pesticides, and carbon from the soil profile as part of the emphasis on erosion

productivity. The models also consider the effect of changes in atmospheric CO_2 on growth and transpiration. They can simulate mixed stands with plant competition for light, water, and nutrients. The multi-subarea capabilities of the APEX model greatly expand the simulation strengths in EPIC. APEX can simulate detailed management practices related to farm animal productions, such as confined or partially confined area feeding, intensive rotational grazing, and cropland grazing after harvest. APEX enables the physically based simulation of conservation practices like filter strips, terraces, and waterways.

The EPIC and APEX models require daily weather data (precipitation, maximum and minimum air temperature, solar radiation, wind speed, and relative humidity). They can be user inputs or generated within the models based on long-term monthly weather statistics. Other main inputs required are soil properties and attributes by layer, land use and field operation schedule, and/or subarea delineation (for APEX).

The model calibration should start with the hydrology component by investigating the PET, ET, crop growth season, and base flow proportion and by comparing simulated flow with observed values. The user can choose to use a different PET equation or curve number equation, and adjust the initial CN2 values for different land use to improve model performance. The crop growth should be realistic to represent proper cover and residue; therefore, if biomass or crop grain yield data are available, they can be calibrated before the calibration of erosion–sedimentation. The user can choose different erosion equations. Calibrations of nutrients and pesticides follow after flow and sediment calibrations. However, because the processes of hydrology, erosion, crop growth, nutrient cycling, and pesticide fate are intimately intertwined, back and forth refining and adjustment may be needed for achieving an overall satisfactory model performance.

In this study the APEX model was manually calibrated and validated for monthly stream flow, sediment yield, and nutrient losses from the North Bosque Watershed at the Hico monitoring station (1993–1998), Texas. The watershed area above Hico is 92,650 ha. It contains 41 dairies. Influential model parameters were adjusted during model calibration. The NSE values ranged from 0.69 to 0.85 for the calibration period. The NSE values ranged from 0.61 to 0.78 for flow, sediment, mineral P, organic P, and total P losses for the validation period. The model performance was relatively poor for organic N and mineral N losses during the validation period. This could partly be attributed to the uncertainty in model input data in the N fertilizer application with the assumption that commercial fertilizer was used for improved pasture (same rate for all the simulation years) and the automatic N fertilizer option was used for cropland, and that no commercial fertilizer was applied for the remaining land uses. No effort was made to

adjust model inputs in regard to numbers of cows and the assignment of waste application fields for individual dairies. However, in general, the model's predictions were reasonable based on statistical criteria for establishing satisfactory water quality model performance proposed by Moriasi et al. (2007). This study shows that the APEX model is a powerful tool for simulating feedlot water quality.

Experience has taught us that combining modeling with data collection programs advances research much more rapidly than either effort alone. The data collection provides essential information for use in developing and testing models, and models can provide feedback to the data collection program that can be used to enhance the experimental design and expand the temporal and spatial scales. In reality, models are often used where data have been collected without knowing the actual model input requirement or where it is not possible to collect all the required inputs, especially for larger scale watershed modeling. Assumptions and best judgment have to be made. With uncertainties associated with model inputs, model parameters, model structures, and observations, it should be expected that model predictions cannot fully replicate observations. Users should not be encouraged to over-tune model inputs.

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